🌊 Coastal Climate Intelligence System

This system combines multi-source coastal data, prediction, anomaly detection, and actionable alerts.

📌 Summary

Location	Data Status	Risk	Age
9.9312, 76.2673	Recent (19 min ago)	SAFE	19 min

Current conditions are SAFE.

🔮 Forecast alert timeline

19:00 — 31% chance of safe conditions in 1 hour(s).

20:00 — 51% chance of safe conditions in 2 hour(s).

21:00 — 56% chance of safe conditions in 3 hour(s).

22:00 — 75% chance of caution conditions in 4 hour(s).

23:00 — 86% chance of caution conditions in 5 hour(s).

00:00 — 94% chance of safe conditions in 6 hour(s).

🧭 Recommendation

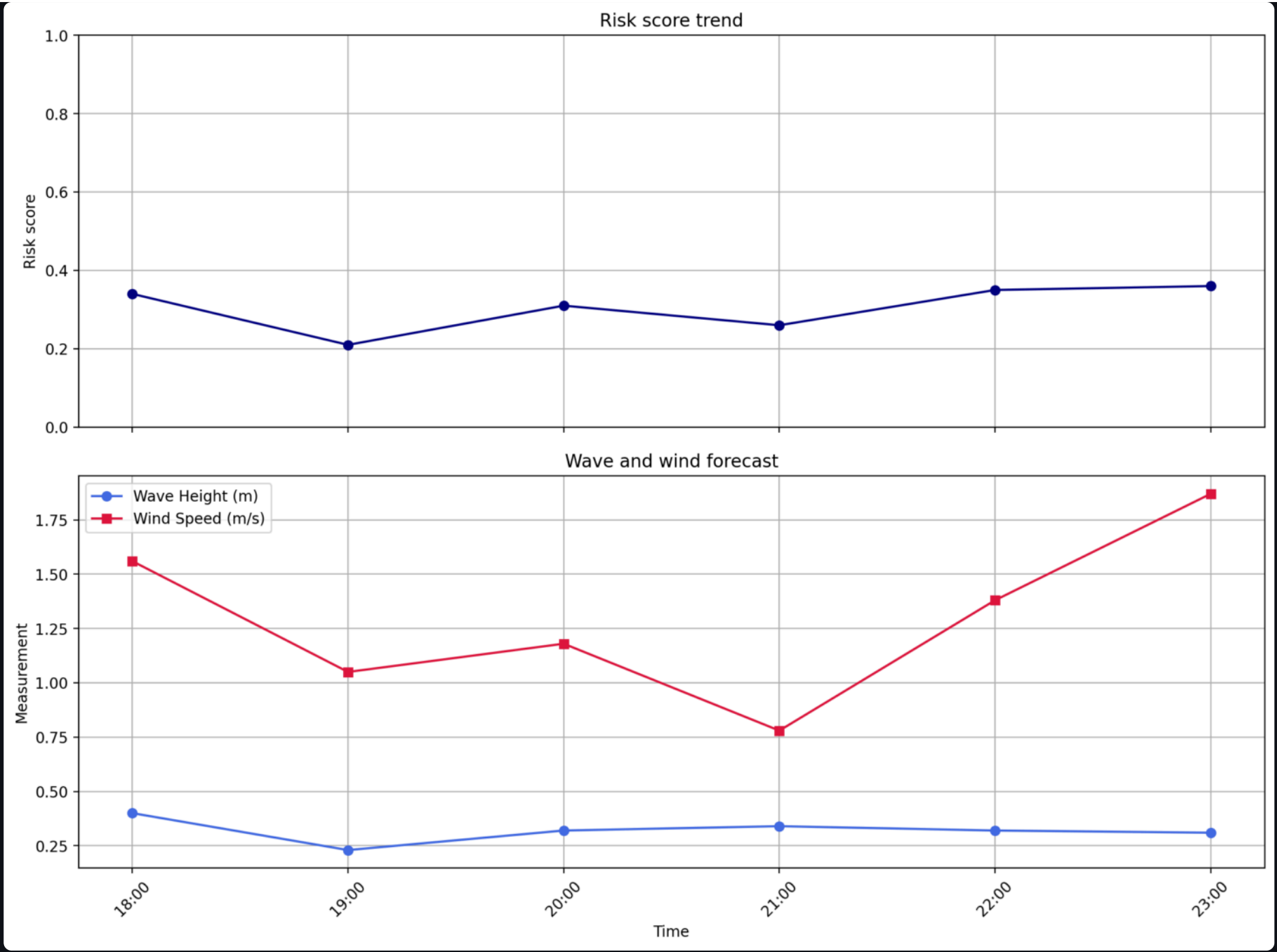
📱 Notification simulation

Avoid sailing and delay trips until conditions improve. Current state: SAFE. normal coastal activity is allowed. Estimated 31% chance of this risk level in the near term.

[SMS] Avoid sailing and delay trips until conditions improve. Current state: SAFE. normal coastal activity is allowed. Estimated 31% chance of this risk level in the near term.

ALERT: SAFE risk for Fishermen. Send this message by voice call or text if the network is weak.

📈 Risk trend and forecast timeline


Risk score trend
Risk score
1.0
0.8
0.6
0.4
0.2
0.0
Wave and wind forecast
Wave Height (m)
Wind Speed (m/s)
Measurement
1.75
1.50
1.25
1.00
0.75
0.50
0.25
18:00
19:00
20:00
21:00
22:00
23:00
Time

📋 Detailed fused risk data

time	waveHeight	windSpeed	riskScore	riskLabel	riskEmoji	riskColor	spike	anomaly
2026-03-27T18:00:00+00:00	0.4	1.56	0.34	SAFE	🟢	green	0	☐
2026-03-27T19:00:00+00:00	0.23	1.05	0.21	SAFE	🟢	green	0	☐
2026-03-27T20:00:00+00:00	0.32	1.18	0.31	SAFE	🟢	green	0.14	☐
2026-03-27T21:00:00+00:00	0.34	0.78	0.26	SAFE	🟢	green	0.02	☐
2026-03-27T22:00:00+00:00	0.32	1.38	0.35	CAUTION	🟡	yellow	0.27	☐
2026-03-27T23:00:00+00:00	0.31	1.87	0.36	CAUTION	🟡	yellow	0.19	☐

🌐 Multi-source fusion

This system blends observations from StormGlass, NOAA, and ECMWF using a weighted average model to improve coastal decision intelligence.